

CSE587: Networks for AI/ML Systems

QUIZ 2 (30 mins) [Total: 18 points]

Sep 4, 2025

Q.1. Imagine you are designing a packet-forwarding application that must sustain 40 Gbps throughput on commodity hardware. What are the key sources (at least two) of overhead in the traditional TCP/IP kernel network stack? Specify **two design changes** to mitigate the mentioned overheads. **[2+2]**

Solution:

- **Overheads:**
 - Data-copy overhead: Multiple packet copies (NIC → kernel buffer → user buffer).
 - Context switching overhead: Per-packet system calls (read/write).
 - Interrupt processing overhead.
 - Per-packet protocol processing overhead (TCP segmentation, IP processing, TCP/IP checksums, etc.).
 - Locking overheads: for shared data structures
- **Design changes:**
 - Use **zero-copy I/O**, i.e., bypass the kernel for packet copy, and directly copy packets into the user space
 - Batch system calls, i.e., GRO/TSO
 - Offload checksum/segmentation to NIC

Q.2. Suppose you plan to use a kernel-bypass solution such as DPDK to achieve 40Gbps speed. You plan to host an HTTP-based web application over the given setup. Will the given solution work as it is? Explain why? Marks will be assigned for correct justification. **[2]**

Solution: NO. HTTP works on top of the TCP protocol, and DPDK does not support TCP. We need stacks such as VPP or mTCP to provide TCP support over DPDK.

Q.3. Suppose you plan to use the following set of optimizations to improve the performance of the traditional Linux kernel. State and explain the scenario (i.e., type of input traffic) when these optimizations will fail to perform, individually. **[4]**

- (a) Generic receive offload (GRO)
- (b) Receive side scaling (RSS) or aRFS (accelerated Receive Flow Steering)

Solution:

- (c) Generic receive offload (GRO): TCP message (or segment) sizes are small, and/or the application is latency-sensitive.
- (d) Receive side scaling (RSS) or aRFS (accelerated Receive Flow Steering): The incoming traffic does not comprise balanced flows. That is, some flows come at very high rates and others at very low rates. This results in an imbalance in CPU core utilization.

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However, overload can be managed by assigning more NIC hardware queues (i.e., more CPUs) to heavy flows.

Q.4. Assume that your deep learning model requires **2048GB** of memory for DNN training. Suppose you have **100 servers with 8 GPUs** each. Each GPU has a capacity of **10GB**. [5]

- How would you train this model with the given infrastructure? [2]
- How much infrastructure is needed (approx)? In terms of the number of servers and GPUs. [2]
- Is there a need to build an efficient networking solution for this use case? Why? [1]

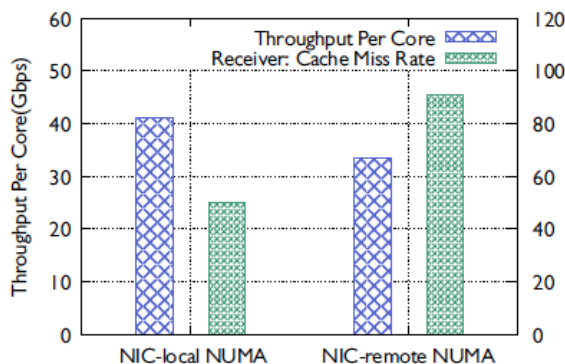
Solution:

Shard the model using Tensor parallelism (TP) or Pipeline parallelism (PP) or both. We will need around 205 GPUs (2048/10) across 26 servers (205/8) to train the model.

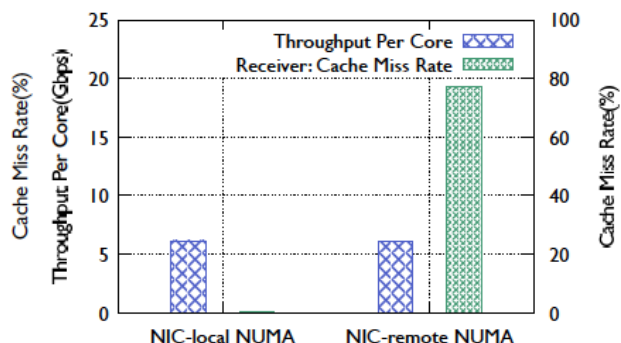
If we implement Data Parallelism (DP), we need to synchronize the gradients across all the GPUs (i.e., 205) during the backward pass. The communication rates are in the order of a few Gigabytes per microsecond, resulting in traffic bursts and causing congestion, which eventually results in performance degradation (i.e., long training times).

However, if we only use TP and PP, network congestion is not a serious problem. TP, typically, involves gradient synchronization within a server; therefore, we need efficient inter-GPU communication within the server. PP requires low-rate communication.

Q.5. The plots below depict the throughput per core and cache miss rate for the traditional network stack, given the flow type and the NUMA node/core. What is your takeaway from this plot? [2]



A) Long flow performance



B) Short flow performance

Solution: Short flow performance does not depend on whether the flow is assigned a CPU core, that is, NIC-local, or NIC-remote, but it does for long flows. This implies that, given a mix of

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short and long flows, we must assign long flows to NIC-local NUMA core and short flows to NIC-remote NUMA core.

Q.6. Which of the following is **NOT** a major overhead in the traditional TCP/IP kernel stack? [1]

- a) System calls
- b) Data copies between kernel and user space
- c) Interrupt handling
- d) Direct memory access (DMA) copy between the NIC and kernel

Solution: d) Direct memory access (DMA) copy between the NIC and kernel
